

WEB SCRAPING for data gathering

Synent Technologies Data Science Internship

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1. Objective

The objective of this project is to extract useful information from a webpage using Python by sending HTTP requests, parsing HTML content, and organizing the extracted data into a structured format for further analysis.

2. Project Information

Project Name: Web Scraping using Python

Source: Public Website

Libraries: Requests, BeautifulSoup (bs4), Pandas

Development Environment: Python, Jupyter Notebook, VS Code

3. Methodology

- Imported required Python libraries.
- Connected to the target webpage using Requests.
- Retrieved HTML source code.
- Parsed HTML using BeautifulSoup.
- Extracted required webpage elements.
- Cleaned and organized extracted data.
- Stored data in a Pandas DataFrame.

4. Results

Successfully extracted structured information from the target webpage.

Key Findings:

- Established connection with website.
- Parsed HTML efficiently.
- Extracted required information.
- Organized data into structured format.

5. Generated Outputs

HTML Parsing Output

Extracted Data Preview

Structured Pandas DataFrame

Cleaned Scraped Dataset

6. Conclusion

This project demonstrates the complete workflow of web scraping using Python, Requests, and BeautifulSoup. The extracted data was cleaned and organized for further analysis, providing a strong foundation for automation and data science applications.